17/2026 Basic Networks Commands

AIM:-

Basic Networking commands in Linux and Windows operating systems.

Basic Networking Commands:-

1) arp -a :- ARP is start form of address resolution protocal, It will show the IP computer along with the IP address and Mac address of your router

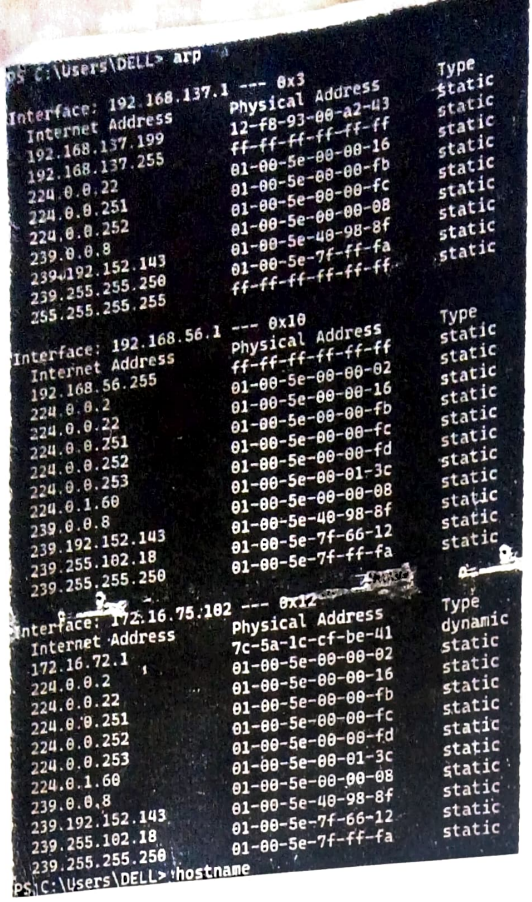
2) host name:. This is the simplest of all TCP/IP Commands. It simply displays complex.

3) ipconfig /all :- This Commands display detailed, Configuration about Connection including Router, Gateway, DNS, DHCP and type of ethernet system.

4) nbstats :- This command helps solve problem which - Netbios name resolution for Netbios over TCP/IP

5) net stat :- (Network Statistics) displays Variety of stat. about a Computer TCP/IP Connections. It is a Command line tool for monitoring network Connection in Com. & Statistics. etc....

Eg:- net stat -v



```
PS C:\Users\DELL> arp

Interface: 192.168.137.1 --- 0x3
  Internet Address      Physical Address      Type
  192.168.137.199       12-f8-93-00-a2-43     static
  192.168.137.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.0.0.8             01-00-5e-40-98-8f     static
  239.192.152.143       01-00-5e-7f-ff-fa     static
  239.255.255.250       ff-ff-ff-ff-ff-ff     static
  255.255.255.255

Interface: 192.168.56.1 --- 0x10
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  224.0.1.60            01-00-5e-00-01-3c     static
  239.0.0.8             01-00-5e-00-00-08     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 172.16.75.102 --- 0x12
  Internet Address      Physical Address      Type
  172.16.72.1           7c-5a-1c-cf-be-41     dynamic
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  224.0.1.60            01-00-5e-00-01-3c     static
  239.0.0.8             01-00-5e-00-00-08     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
PS C:\Users\DELL> hostname
```

```
C:\Users\DELL>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
  -g host-list      Loose source route along host-list.
  -h maximum_hops   Maximum number of hops to search for target.
  -i address        Use the specified source address.
  -n                Do not resolve addresses to hostnames.
  -p period         Wait period milliseconds between pings.
  -q num_queries    Number of queries per hop.
  -w timeout        Wait timeout milliseconds for each reply.
  -4                Force using IPv4.
  -6                Force using IPv6.
```

```
PS C:\Users\DELL> hostname
HDC0124162
PS C:\Users\DELL> ipconfig / all

Error: unrecognized or incomplete command line.

USAGE:
    ipconfig [/allcompartments] [/? | /all |
                                 /renew [adapter] | /release [adapter] |
                                 /renew6 [adapter] | /release6 [adapter] |
                                 /flushdns | /displaydns | /registerdns |
                                 /showclassid adapter |
                                 /setclassid adapter [classid] |
                                 /showclassid6 adapter |
                                 /setclassid6 adapter [classid] ]


where
    adapter             Connection name
                        (wildcard characters * and ? allowed, see examples)

    Options:
       /?               Display this help message
       /all             Display full configuration information.
       /release         Release the IPv4 address for the specified adapter.
       /release6        Release the IPv6 address for the specified adapter.
       /renew           Renew the IPv4 address for the specified adapter.
       /renew6          Renew the IPv6 address for the specified adapter.
       /flushdns        Purges the DNS Resolver cache.
       /registerdns     Refreshes all DHCP leases and re-registers DNS names
       /displaydns      Display the contents of the DNS Resolver Cache.
       /showclassid     Displays all the dhcp class IDs allowed for adapter.
       /setclassid      Modifies the dhcp class id.
       /showclassid6    Displays all the IPv6 DHCP class IDs allowed for adapter.
       /setclassid6     Modifies the IPv6 DHCP class id.


The default is to display only the IP address, subnet mask and
default gateway for each adapter bound to TCP/IP.

For Release and Renew, if no adapter name is specified, then the IP address
leases for all adapters bound to TCP/IP will be released or renewed.

For Setclassid and Setclassid6, if no ClassId is specified, then the ClassId is remov

Examples:
    > ipconfig                  ... Show information
    > ipconfig /all             ... Show detailed information
    > ipconfig /renew           ... renew all adapters
    > ipconfig /renew EL*       ... renew any connection that has its
                                    name starting with EL
    > ipconfig /release *Con*
```

6) NS look up :- (Name server look up) is a tool used to perform DNS lookups in Linux domain DNS details, such as the IP address of a particular computer, the MX domain or the NS Servers of a domain. ns lookup can operate in two modes :- non-interactive of interactive

eg :- ns look up www.google.com

7) Path ping :- Pathping is unique to window's and is basically a combination of Tracert Commands. Pathping forces the route to the description address then second test of each router along the way, gathering Statistics on the rate of drathop.

8) Ping :- (Pocket Internet Groper) Command is the best way to test connective nodes. Ping Use ICMP (Internet Control Message Protocol) to Communicate.

1. # Ping hostname (Ping localhost)
2. # Ping ip address (Ping 4.2.2.2)
3. # Ping fully qualified domain name.

9) Route :- Route Command is used to show/ manipulate the IP routing table. It is to setup static routes to specific host or network via an interface

```
PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.

PS C:\Users\DELL> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:52955        HDC0124162:52956       ESTABLISHED
  TCP    127.0.0.1:52956        HDC0124162:52955       ESTABLISHED
  TCP    127.0.0.1:52957        HDC0124162:52958       ESTABLISHED
  TCP    127.0.0.1:52958        HDC0124162:52957       ESTABLISHED
  TCP    172.16.75.102:55921    hkg12s32-in-f10:https  TIME_WAIT
  TCP    172.16.75.102:56167    lcmaaa-ax-in-f10:https TIME_WAIT
  TCP    172.16.75.102:58804    13.89.179.12:https     ESTABLISHED
  TCP    172.16.75.102:58807    52.123.226.20:https    ESTABLISHED
  TCP    172.16.75.102:58813    lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.75.102:60807    4.213.25.242:https     ESTABLISHED
  TCP    172.16.75.102:61755    pnmaaa-aq-in-f5:https  TIME_WAIT
  TCP    172.16.75.102:64029    cgk03s04-in-f5:https   ESTABLISHED
PS C:\Users\DELL>

    -f                Set Don't Fragment flag in packet (IPv4-only).
    -i TTL            Time To Live.
    -v TOS            Type Of Service (IPv4-only. This setting has been deprecated
                      and has no effect on the type of service field in the IP
                      Header).
    -r count          Record route for count hops (IPv4-only).
    -s count          Timestamp for count hops (IPv4-only).
    -j host-list      Loose source route along host-list (IPv4-only).
    -k host-list      Strict source route along host-list (IPv4-only).
    -w timeout        Timeout in milliseconds to wait for each reply.
    -R                Use routing header to test reverse route also (IPv6-only).
                      Per RFC 5095 the use of this routing header has been
                      deprecated. Some systems may drop echo requests if
                      this header is used.
    -S srcaddr        Source address to use.
    -c compartment    Routing compartment identifier.
    -p                Ping a Hyper-V Network Virtualization provider address.
    -4                Force using IPv4.
    -6                Force using IPv6.

PS C:\Users\DELL> nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```